
ASSIGNMENT

PROGRAMMING USING C

WEEK – 07

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DEPT & SEC :ECE F

Week-7

Sunny and Johnny like to pool their money and go to the ice cream parlor. Johnny never buys the same flavor that Sunny does. The only other rule they have is that they spend all of their money.

```
#include<stdio.h>
int main()
{
    int t,m,n,c=0;
    scanf("%d",&t);
    for(int i=0;i<t;i++)
    {
        c=0;
        scanf("%d\n%d",&m,&n);
        int arr[n];
        for(int j=0;j<n;j++)
        {
            scanf("%d",&arr[j]);
        }
        for(int a=0;a<n-1;a++)
        {
            for(int b=a+1;b<n;b++)
            {
                if(arr[a]+arr[b]==m)
                {
                    printf("%d %d\n",a+1,b+1);
                    c=1;
                    break;
                }
            }
            if(c==1)
                break;
        }
    }
}
```

	Input	Expected	Got	
✓	2 4 5 1 4 5 3 2 4 4 2 2 4 3	1 4 1 2	1 4 1 2	✓

Passed all tests! ✓

Numeros the Artist had two lists that were permutations of one another. He was very proud. Unfortunately, while transporting them from one exhibition to another, some numbers were lost out of the first list. Can you find the missing numbers?

```
#include<stdio.h>
int main()
{
    int n,m,c,c1=0,co;
    scanf("%d",&n);
    int arr[n];
    for(int a=0;a<n;a++)
    {
        scanf("%d",&arr[a]);
    }
    scanf("%d",&m);
    int brr[m],ans[m];
    for(int b=0;b<m;b++)
    {
        scanf("%d",&brr[b]);
    }
    for(int j=0;j<m;j++)
    {
        c=0;
        for(int i=0;i<n;i++)
        {
            if(arr[i]==brr[j])
            {
                c=1;
                arr[i]=-1;
                break;
            }
        }
        if(c==0)
        {
            ans[c1]=brr[j];
            c1++;
        }
    }
    for(int a=0;a<c1;a++)
    {
        co=0;
        for(int b=0;b<c1;b++)
        {
            if(ans[b]<ans[a])
                co++;
        }
        int temp=ans[a];
        ans[a]=ans[co];
        ans[co]=temp;
    }
    for(int i=0;i<c1;i++)
        printf("%d ",ans[i]);
    return 0;
}
```

	Input	Expected	Got	
✓	10 203 204 205 206 207 208 203 204 205 206 13 203 204 204 205 206 207 205 208 203 206 205 206 204	204 205 206	204 205 206	✓

Passed all tests! ✓

Watson gives Sherlock an array of integers. His challenge is to find an element of the array such that the sum of all elements to the left is equal to the sum of all elements to the right. For instance, given the array **arr = [5, 6, 8, 11]**, 8 is between two subarrays that sum to **11**. If your starting array is **[1]**, that element satisfies the rule as left and right sum to **0**.

```
#include<stdio.h>
int main()
{
    int t,n,is,rs,m;
    scanf("%d",&t);
    for(int i=0;i<t;i++)
    {
        is=0;
        rs=0;
        scanf("%d",&n);
        int arr[n];
        for(int j=0;j<n;j++)
        {
            scanf("%d",&arr[j]);
        }
        m=n/2;
        if(arr[m]==0)

            for(m=0;arr[m]==0&&m<n;m++);

        for(int j=0;j<=m;j++)
            is=is+arr[j];
        for(int j=m;j<n;j++)
            rs=rs+arr[j];
        printf("%s\n",(is==rs)?"YES":"NO");
    }
    return 0;
}
```

	Input	Expected	Got	
✓	3 5 1 1 4 1 1 4 2 0 0 0 4 0 0 2 0	YES YES YES	YES YES YES	✓
✓	2 3 1 2 3 4 1 2 3 3	NO YES	NO YES	✓

Passed all tests! ✓